

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

In the Matter of CERTAIN MOBILE ELECTRONIC DEVICES AND COMPONENTS THEREOF	Investigation No. 337-TA-_____
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**COMPLAINT UNDER SECTION 337 OF THE
TARIFF ACT OF 1930, AS AMENDED**

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EXHIBIT LIST

Exhibits	Description
1	Copy of U.S. Patent No. 12,530,925
2	Certified Assignment Records for U.S. Patent No. 12,530,925
3	Copy of U.S. Patent No. 12,288,421
4	Certified Assignment Records for U.S. Patent No. 12,288,421
5	Copy of U.S. Patent No. 12,456,334
6	Certified Assignment Records for U.S. Patent No. 12,456,334
7	Copy of U.S. Patent No. 12,361,760
8	Assignment Records for U.S. Patent No. 12,361,760
9	Copy of U.S. Patent No. 12,298,394
10	Certified Assignment Records for U.S. Patent No. 12,298,394
11	Copy of U.S. Patent No. 12,332,352
12	Assignment Records for U.S. Patent No. 12,332,352
13	<i>The Future Is Here: iPhone X</i> , Apple (Sept. 12, 2017), available at https://www.apple.com/newsroom/2017/09/the-future-is-here-iphone-x/
14	Raymond Wong, <i>Hackers Fool the iPhone X's Face ID Using a Cheesy-Looking Mask</i> , Mashable (Nov. 13, 2017), available at https://mashable.com/article/apple-iphone-x-face-id-mask-hack
15	Daniel Oberhaus, <i>iPhone X's Face ID Can Be Fooled with a 3D-Printed Mask</i> , Vice (Nov. 13, 2017), available at https://www.vice.com/en/article/iphone-x-face-id-mask-spoof/
16	List of Foreign Counterparts
17	U.S. Patent No. 12,530,925 Infringement Chart
18	U.S. Patent No. 12,288,421 Infringement Chart
19	U.S. Patent No. 12,456,334 Infringement Chart
20	U.S. Patent No. 12,361,760 Infringement Chart
21	U.S. Patent No. 12,298,394 Infringement Chart
22	U.S. Patent No. 12,332,352 Infringement Chart
23	Collection of Apple Manuals and Data Sheets
24	Evidence of Importation
25	Public Version of Domestic Industry Claim Chart for 12,530,925
25C	Confidential Domestic Industry Claim Chart for 12,530,925
26	Public Version of Domestic Industry Claim Chart for 12,288,421
26C	Confidential Domestic Industry Claim Chart for 12,288,421
27	Public Version of Domestic Industry Claim Chart for 12,456,334
27C	Confidential Domestic Industry Claim Chart for 12,456,334
28	Public Version of Domestic Industry Claim Chart for 12,361,760
28C	Confidential Domestic Industry Claim Chart for 12,361,760
29	Public Version of Domestic Industry Claim Chart for 12,298,394
29C	Confidential Domestic Industry Claim Chart for 12,298,394
30	Public Version of Domestic Industry Claim Chart for 12,332,352
30C	Confidential Domestic Industry Claim Chart for 12,332,352
31	Public Version of Declaration of Uwe Schneider

Exhibits	Description
31C	Confidential Declaration of Uwe Schneider
32	<i>Apple iPhone 17 Homepage</i> , Apple, available at https://www.apple.com/iphone-17/ (last accessed Aug. 10, 2026)
33	<i>Apple iPhone 17 Specifications</i> , Apple, available at https://www.apple.com/iphone-17/specs/ (last accessed Aug. 10, 2026)
34	<i>Apple iPhone 17 Face ID: Technology, Process & Cost Report</i> , YOLE Group (2026) (Public Version)
34C	<i>Apple iPhone 17 Face ID: Technology, Process & Cost Report</i> , YOLE Group (2026)
35	<i>Apple Platform Security</i> , Apple (Dec. 2024), available at https://web.archive.org/web/20250927020819/https://help.apple.com/pdf/security/en_US/apple-platform-security-guide.pdf
36	<i>About Face ID Advanced Technology</i> , Apple (Dec. 9, 2024), available at https://support.apple.com/en-us/102381
37	<i>Product Systems Overview</i> , trinamiX (Aug. 6, 2026) (Public Version)
37C	<i>Product Systems Overview</i> , trinamiX (Aug. 6, 2026)
38	<i>Technical Datasheet: trinamiX Face Authentication Evaluation Kit ET 90</i> , trinamiX (Nov. 2022) (Public Version)
38C	<i>Technical Datasheet: trinamiX Face Authentication Evaluation Kit ET 90</i> , trinamiX (Nov. 2022)
39	<i>User Manual: trinamiX Face Authentication Evaluation Kit ET 90</i> , trinamiX (Nov. 2022) (Public Version)
39C	<i>User Manual: trinamiX Face Authentication Evaluation Kit ET 90</i> , trinamiX (Nov. 2022)
40	<i>ET 90 Specification</i> , trinamiX (Apr. 8, 2022) (Public Version)
40C	<i>ET 90 Specification</i> , trinamiX (Apr. 8, 2022)
41	<i>Cockatiel System Specification</i> , trinamiX (Apr. 30, 2025) (Public Version)
41C	<i>Cockatiel System Specification</i> , trinamiX (Apr. 30, 2025)
42	<i>BlueBird System Specification</i> , trinamiX (June 2, 2022) (Public Version)
42C	<i>BlueBird System Specification</i> , trinamiX (June 2, 2022)
43	<i>Silver Eagle System Specification</i> , trinamiX (Aug. 26, 2024) (Public Version)
43C	<i>Silver Eagle System Specification</i> , trinamiX (Aug. 26, 2024)
44	<i>Changelight Product Specification</i> , Changelight (Public Version)
44C	<i>Changelight Product Specification</i> , Changelight
45	<i>HDK8450 (SM8450P) Processor: User Guide</i> , Qualcomm (Dec. 2021), available at https://www.lantronix.com/wp-content/uploads/pdf/PMD-00133A_hdk8450_sm8450p_processor_user_guide-3.pdf
46	<i>NILT Lens Specification</i> , trinamiX (Jan. 31, 2024) (Public Version)
46C	<i>NILT Lens Specification</i> , trinamiX (Jan. 31, 2024)
47	Expert Declaration of Dr. Ioannis Kakadiaris

APPENDIX LIST

Appendices	Description
A	Certified Prosecution History of U.S. Patent No. 12,530,925
B	Certified Prosecution History of U.S. Patent No. 12,288,421
C	Certified Prosecution History of U.S. Patent No. 12,456,334
D	Prosecution History of U.S. Patent No. 12,361,760
E	Certified Prosecution History of U.S. Patent No. 12,298,394
F	Prosecution History of U.S. Patent No. 12,332,352

I. INTRODUCTION

1. Complainants trinamiX Sensing LLC and trinamiX GmbH (“Complainants” or “trinamiX”) respectfully file this Complaint under Section 337 of the Tariff Act of 1930, as amended, 19 U.S.C. § 1337, to stop the unlawful importation into the United States, sale for importation into the United States, offering for sale, and/or sale within the United States after importation of certain iPhones and iPads incorporating face authentication, 3D-sensing, and/or material detection capabilities that infringe valid and enforceable patents owned by trinamiX GmbH by assignment.

2. The proposed Respondent is Apple Inc. (“Respondent” or “Apple”), which is headquartered in Cupertino, California.

3. This Complaint is based on Respondent’s unlawful importation into the United States, sale for importation into the United States, offering for sale, and/or sale within the United States after importation of certain iPhones and iPads incorporating face authentication, 3D-sensing, and/or material detection capabilities that infringe one or more claims of the following trinamiX patents (the “Asserted Patents”), either literally or under the doctrine of equivalents:

Asserted Patent	Claims Asserted Against Respondent (independent claims in bold)
12,530,925	1 , 5, 6, 7, 8, 17 , 21, 22, 23, 24
12,288,421	1 , 5, 6, 7, 8, 9, 11, 17, 20, 21, 23, 24
12,456,334	1 , 7, 11, 13, 15
12,361,760	1 , 4, 5, 6, 8, 9, 10, 13, 15
12,298,394	1 , 5, 6, 15 , 18, 19
12,332,352	1 , 2, 9, 14, 16, 23, 28

4. Copies of the Asserted Patents are attached as Exhibits 1, 3, 5, 7, 9, and 11.¹ trinamiX GmbH owns by assignment the entire right, title, and interest in and to the Asserted Patents, as reflected in the recorded assignments attached as Exhibits 2, 4, 6, 8, 10, and 12.²

5. Pursuant to Commission Rules 210.12(a)(12) and 210.10(b)(1), the accused products are Respondent's iPhones and iPads and components thereof, incorporating face authentication, 3D-sensing, and/or material detection capabilities (hereinafter, individually and collectively, "Accused Products").

6. To stop Respondent's unlawful acts, trinamiX requests a limited exclusion order under 19 U.S.C. § 1337(d)(1) barring from entry into the United States all Accused Products and any other Apple iPhones and/or iPads that infringe one or more Asserted Patents and that are or will be imported into the United States, sold for importation into the United States, offered for sale, and/or sold in the United States after importation by or on behalf of Respondent.

7. trinamiX also seeks permanent cease and desist orders under 19 U.S.C. § 1337(f) prohibiting Respondent and its domestic subsidiaries, related companies, affiliates, successors, assigns, and agents from importing, selling for importation, marketing, distributing, offering for sale, selling after importation, or otherwise transferring (except for exportation) within the United States all Accused Products and any other Apple iPhones and/or iPads that infringe one or more Asserted Patents.

¹ trinamiX has ordered but not yet received certified copies of the Asserted Patents from the USPTO. trinamiX will provide the certified copies as soon as trinamiX receives them.

² trinamiX has ordered but not yet received certified copies of the assignment records and prosecution history for the '760 and '352 patents from the USPTO. trinamiX will provide the certified copies as soon as trinamiX receives them.

8. trinamiX further seeks the imposition of a bond upon importation of all Accused Products and any other Apple iPhones and/or iPads that infringe one or more Asserted Patents during the 60-day Presidential review period pursuant to 19 U.S.C. § 1337(j).

9. As required by Section 337(a)(2) and defined by Section 337(a)(3), an industry in the United States relating to articles protected by the Asserted Patents exists, or alternatively is in the process of being established, as set forth in Section X of this Complaint. This showing rests on trinamiX's significant and substantial investments in its products and technologies that are protected by the Asserted Patents.

II. COMPLAINANTS

10. trinamiX GmbH is a limited liability company organized and existing under the laws of the Federal Republic of Germany, with its principal place of business at Industriestraße 35, 67063 Ludwigshafen am Rhein, Germany.

11. trinamiX Sensing LLC is a limited liability company organized and existing under the laws of the State of Delaware, and is registered to transact business in Texas, with its principal place of business at 7800 Shoal Creek Boulevard, Suite 200N, Austin, Texas 78757.

12. trinamiX originated from research at BASF SE ("BASF"), a chemical company headquartered in Ludwigshafen am Rhein, Germany. Beginning in or around 2010, BASF scientists working on organic solar cells identified an unexpected material effect with applications far beyond that original context, including sensing and three-dimensional measurement. That discovery led trinamiX's founding scientists to build early three-dimensional camera prototypes and to file patent applications directed to the underlying technology.

13. trinamiX GmbH was formed as a legal entity on December 16, 2014, and commenced operations in 2015, when BASF spun the three-dimensional camera project off into a

dedicated company headquartered in Ludwigshafen am Rhein, Germany, to develop advanced 3D and material sensing technologies. trinamiX GmbH remains a wholly owned subsidiary of BASF.

14. In the decade since, trinamiX has grown from a startup into an operating company that designs, develops, and commercializes advanced sensing technologies, including biometric imaging, three-dimensional vision, and near-infrared spectroscopy solutions for consumer and industrial applications. trinamiX's products are directed to improving decision-making, safety, and product authentication, and trinamiX offers them to customers in the consumer electronics, automotive, and access control markets.

15. trinamiX's patented face authentication technologies are the result of that development work. They add an additional security layer to face authentication by classifying between different materials. An illumination unit captures not only a 2D image and 3D depth information of the face in front of the device, but further differentiates between real skin and other materials. Spoofing the system with a lifelike mask, 3D sculpture, or 2D printout becomes incredibly difficult. As described in Section IV below, those technologies have been tested and certified under the FIDO Biometric Component Certification Program, the IIFAA Biometric Face Security Test Requirement, and the Android Biometric Security Test. trinamiX has supplied face authentication evaluation kits to, and conducted development projects with, prospective customers evaluating trinamiX's face authentication technologies for integration into their own products.

16. trinamiX conducts a growing share of that development work in the United States. trinamiX GmbH holds more than 800 granted or pending patents worldwide, including the Asserted Patents.

17. trinamiX maintains a U.S. presence through trinamiX Sensing LLC, which engages in the innovation, research, design, and development of trinamiX's face authentication products

and technologies in the United States that embody the Asserted Patents where it employs dedicated specialists in software engineering and development, hardware engineering, and data science at its Austin, Texas facility. trinamiX's U.S. investments in plant and equipment, labor and capital, and engineering and research and development are described *infra* in Section X.B. and in the Confidential Declaration of Uwe Schneider (Confidential Exhibit 31C).

18. trinamiX's patents, including the Asserted Patents, arose from trinamiX's own research and development. trinamiX's patents protect products that trinamiX designs, develops, and offers to customers. trinamiX does not hold or assert its patents as a non-practicing licensing business. trinamiX seeks relief from Apple's unfair trade practices to protect an industry it has built and is continuing to build in the United States.

III. RESPONDENT

19. Apple is a U.S. corporation having its principal place of business at One Apple Park Way, Cupertino, California 95014. Apple unlawfully sells for importation, imports, and/or sells after importation into the United States certain iPhones and iPads incorporating face authentication, 3D-sensing, and/or material detection capabilities, including the Accused Products, that infringe the Asserted Patents, either literally or under the doctrine of equivalents.

20. Apple is in the business of designing, manufacturing, and marketing smartphones, personal computers, tablets, wearables, and accessories, and sells a variety of related services. As discussed above, the Accused Products are certain iPhone and iPad devices incorporating face authentication, 3D-sensing, and/or material detection capabilities.

IV. THE TECHNOLOGY AND PRODUCTS AT ISSUE

A. Background of the Technology

21. One of trinamiX's core technologies is its face authentication system, which provides a critical layer of security beyond conventional face recognition through trinamiX's

patent-protected material and skin classification technologies (hereinafter, “trinamiX’s face authentication technologies”). trinamiX’s face authentication technologies are designed for integration across consumer electronics, automotive, and access control applications.

22. trinamiX’s face authentication technologies were designed to address a well-documented security gap in conventional face authentication technology. Conventional face recognition systems are vulnerable to spoofing—researchers and journalists have repeatedly demonstrated that many face unlock features can be defeated using photographs, 3D-printed masks, or silicone replicas. The material and skin detection technologies forming a part of trinamiX’s face authentication technologies address these vulnerabilities by providing specific methods of verifying that the face presented to a device belongs to a real human, and not a realistic mask, photograph, or other spoof.

23. In 2016 and 2017, trinamiX scientists developed a proprietary beam profile analysis technique, a method that projects an illumination pattern comprising a plurality of illumination features onto a scene and analyzes the reflection features generated by the scene in response to the illumination to determine properties of the illuminated surface. In plain terms, the technique projects a pattern of light onto a scene and analyzes how that light reflects back to determine object properties.

24. Over time, trinamiX recognized that its proprietary beam profile analysis technique could be applied to improve the security of conventional face authentication systems. trinamiX thus developed face authentication technologies that not only verify a face is present in the scene, but also determine whether the material of the detected face is human skin based on the properties of the reflected light, and verify that the face exhibits three-dimensional depth consistent with real

facial characteristics. trinamiX's face authentication technologies provide a powerful defense against spoofing attacks using photographs, masks, and other non-skin materials.

25. trinamiX's face authentication technologies have been tested and certified under the FIDO Biometric Component Certification Program, the IIFAA Biometric Face Security Test Requirement, and the Android Biometric Security Test, which are well-recognized industry tests for biometric security measures. trinamiX's face authentication technologies have passed these certifications with a Spoof Acceptance Rate of 0% after exposure to spoof attempts, including ultra-realistic masks, over a period of many weeks.

B. Products at Issue

26. The Accused Products are iPhones and iPads and components thereof incorporating face authentication, 3D-sensing, and/or material detection capabilities. Apple first released face authentication, called "Face ID," on or around November 3, 2017, with the iPhone X, and added it to the iPad the following year. Exhibit 13, at 2 (*The Future Is Here: iPhone X*, Apple (Sept. 12, 2017), available at <https://www.apple.com/newsroom/2017/09/the-future-is-here-iphone-x/>).

27. On information and belief, the original release of Face ID did not incorporate trinamiX's face authentication technologies covered by the Asserted Patents and lacked material and skin detection capabilities. Indeed, technical publications report that the Face ID of the iPhone X was able to be tricked using simply constructed masks. Exhibit 14, at 1 ("Security researchers at Bkav have reportedly defeated the iPhone X's Face ID feature using a simply-constructed 3D mask.") (Raymond Wong, *Hackers Fool the iPhone X's Face ID Using a Cheesy-Looking Mask*, Mashable (Nov. 13, 2017), available at <https://mashable.com/article/apple-iphone-x-face-id-mask-hack>); Exhibit 15, at 1 ("[R]esearchers at the Vietnamese cybersecurity firm Bkav announced that they had made a mask that could fool the iPhone X's . . . Face ID. The mask consists of a 3D-printed frame, a silicone nose that was hand-sculpted by artists, and a few photos that were layered

on top of the mask.”) (Daniel Oberhaus, *iPhone X’s Face ID Can Be Fooled with a 3D-Printed Mask*, Vice (Nov. 13, 2017), available at <https://www.vice.com/en/article/iphone-x-face-id-mask-spoof/>). Following the introduction of the Face ID feature, Apple continued to develop it and later incorporated the patented trinamiX face authentication technologies in the Accused Products.

V. THE ASSERTED PATENTS AND NON-TECHNICAL DESCRIPTIONS OF THE INVENTIONS

28. trinamiX asserts six patents in this Complaint: U.S. Patent No. 12,530,925 (“the ’925 patent”), U.S. Patent No. 12,288,421 (“the ’421 patent”), U.S. Patent No. 12,456,334 (“the ’334 patent”), U.S. Patent No. 12,361,760 (“the ’760 patent”), U.S. Patent No. 12,298,394 (“the ’394 patent”), and U.S. Patent No. 12,332,352 (“the ’352 patent”). These patents are briefly described below.

A. The ’925 Patent

1. Identification and Ownership of the ’925 Patent

29. trinamiX GmbH owns by assignment the entire right, title, and interest in and to the ’925 patent, titled “Optical skin detection for face unlock,” which was duly issued by the United States Patent and Trademark Office on January 20, 2026, and names Peter Schillen, Benjamin Guthier, Friedrich Schick, Manuel Guenther, Lars Diesselberg, and Christian Lennartz as co-inventors. The ’925 patent issued from U.S. Patent Application Serial No. 19/088,110, filed on March 24, 2025. The application is a continuation application of U.S. Patent Application Ser. No. 18/468,352, filed Sep. 15, 2023, which is a continuation application of U.S. Patent Application Ser. No. 18/264,326, filed Aug. 4, 2023, which is a U.S. National Phase Application of International Patent Application No. PCT/EP22/053941, filed Feb. 17, 2022, which claims priority to EP patent application Ser. No. 21/157,800.0, filed Feb. 18, 2021, the entire contents of which

are hereby incorporated by reference herein. The expiration date of the '925 patent is February 17, 2042.

30. A copy of the '925 patent is attached as Exhibit 1. A certified copy of the assignment from the named inventors to trinamiX GmbH is attached as Exhibit 2. A certified copy of the prosecution history of the '925 patent is included as Appendix A.

2. Foreign Counterparts to the '925 Patent

31. Exhibit 16 lists each foreign patent and each pending foreign patent application (not already issued as a patent), and each foreign patent application that has been denied, abandoned, or withdrawn, corresponding to the '925 patent, with an indication of the prosecution status of each such patent application. No other foreign patents or patent applications corresponding to the '925 patent have been filed, abandoned, withdrawn, or rejected.

3. Non-Technical Description of the '925 Patent

32. Mobile and other electronic devices may incorporate biometric authentication techniques, such as face authentication. Such face authentication must be secure and be able to protect against attempts to unlock the device by unauthorized persons. Such persons may employ various spoofing techniques that attempt to mimic an authorized user's face, including photographs, masks, or other spoofs that resemble a user's face. It is thus desirable to provide a face authentication technique that can provide high levels of security and protection against a wide variety of spoofs.

33. The '925 patent is directed to systems and methods for securely authenticating a user's face by determining whether the face presented to the device is human skin rather than a spoof. The patented system uses a near-infrared camera and an illumination unit to project a pattern of near-infrared light onto the user's face, captures images of the light reflected from the face, and analyzes the reflected light patterns to make two independent security determinations: first, a

material classification that compares the intensity distributions of the reflected light against reference data to determine whether the surface is human skin as opposed to paper, plastic, silicone, or other non-biological materials; and second, a depth analysis that determines whether the face has three-dimensional geometry consistent with a human face rather than the flat profile of a photograph or display. Authentication is granted only if both the material classification confirms real skin and the depth analysis confirms a three-dimensional object.

B. The '421 Patent

1. Identification and Ownership of the '421 Patent

34. trinamiX GmbH owns by assignment the entire right, title, and interest in and to the '421 patent, titled "Optical skin detection for face unlock," which was duly issued by the United States Patent and Trademark Office on April 29, 2025, and names Peter Schillen, Benjamin Guthier, Friedrich Schick, Manuel Guenther, Lars Diesselberg, and Christian Lennartz as co-inventors. The '421 patent issued from U.S. Patent Application Serial No. 18/743,640, filed on June 14, 2024. The application is a continuation application of U.S. Patent Application Ser. No. 18/468,352, filed Sep. 15, 2023, which is a continuation application of U.S. Patent Application Ser. No. 18/264,326, filed Aug. 4, 2023, which is a U.S. National Phase Application of International Patent Application No. PCT/EP22/053941, filed Feb. 17, 2022, which claims priority to EP patent application Ser. No. 21/157,800.0, filed Feb. 18, 2021, the entire contents of which are hereby incorporated by reference herein. The expiration date of the '421 patent is February 17, 2042.

35. A copy of the '421 patent is attached as Exhibit 3. A certified copy of the assignment from the named inventors to trinamiX GmbH is attached as Exhibit 4. A certified copy of the prosecution history of the '421 patent is included as Appendix B.

2. Foreign Counterparts to the '421 Patent

36. Exhibit 16 lists each foreign patent and each pending foreign patent application (not already issued as a patent), and each foreign patent application that has been denied, abandoned, or withdrawn, corresponding to the '421 patent, with an indication of the prosecution status of each such patent application. No other foreign patents or patent applications corresponding to the '421 patent have been filed, abandoned, withdrawn, or rejected.

3. Non-Technical Description of the '421 Patent

37. Mobile and other electronic devices may incorporate biometric authentication techniques, such as face authentication. Such face authentication must be secure and be able to protect against attempts to unlock the device by unauthorized persons. Such persons may employ various spoofing techniques that attempt to mimic an authorized user's face, including photographs, masks, or other spoofs that resemble a user's face. It is thus desirable to provide a face authentication technique that can provide high levels of security and protection against a wide variety of spoofs.

38. The '421 patent is directed to a method for face authentication that determines whether a face presented to a device is human skin with three-dimensional geometry, rather than a spoof such as a photograph, mask, or replica. The method projects an illumination pattern—such as a point pattern, line pattern, stripe pattern, checkerboard pattern, or other periodic or non-periodic arrangement—onto the scene and captures an image of the light reflected from the face. The system performs independent checks, including a skin detection step that analyzes the material properties of the reflected light to determine whether the surface is human skin; a 3D detection step that uses structured light techniques to determine depth information, verifying that the face has three-dimensional geometry; and at least one authentication step that comprises authenticating

the detected face if the detected face is characterized as skin and the detected face is characterized as a three-dimensional object.

C. The '334 Patent

1. Identification and Ownership of the '334 Patent

39. trinamiX GmbH owns by assignment the entire right, title, and interest in and to the '334 patent, titled "Optical skin detection for face unlock," which was duly issued by the United States Patent and Trademark Office on October 28, 2025, and names Peter Schillen, Benjamin Guthier, Friedrich Schick, Manuel Guenther, Lars Diesselberg, and Christian Lennartz as co-inventors. The '334 patent issued from U.S. Patent Application Serial No. 18/468,352, filed on September 15, 2023. The application is a continuation application of U.S. Patent Application Ser. No. 18/264,326, filed Aug. 4, 2023, which is a U.S. National Phase Application of International Patent Application No. PCT/EP22/053941, filed Feb. 17, 2022, which claims priority to EP patent application Ser. No. 21/157,800.0, filed Feb. 18, 2021, the entire contents of which are hereby incorporated by reference herein. The expiration date of the '334 patent is February 17, 2042.

40. A copy of the '334 patent is attached as Exhibit 5. A certified copy of the assignment from the named inventors to trinamiX GmbH is attached as Exhibit 6. A certified copy of the prosecution history of the '334 patent is included as Appendix C.

2. Foreign Counterparts to the '334 Patent

41. Exhibit 16 lists each foreign patent and each pending foreign patent application (not already issued as a patent), and each foreign patent application that has been denied, abandoned, or withdrawn, corresponding to the '334 patent, with an indication of the prosecution status of each such patent application. No other foreign patents or patent applications corresponding to the '334 patent have been filed, abandoned, withdrawn, or rejected.

3. Non-Technical Description of the '334 Patent

42. Mobile and other electronic devices may incorporate biometric authentication techniques, such as face authentication. Such face authentication must be secure and be able to protect against attempts to unlock the device by unauthorized persons. Such persons may employ various spoofing techniques that attempt to mimic an authorized user's face, including photographs, masks, or other spoofs that resemble a user's face. It is thus desirable to provide a face authentication technique that can provide high levels of security and protection against a wide variety of spoofs.

43. The '334 patent is directed to a face authentication system that uses a near-infrared (NIR) camera that is sensitive to NIR light in the range of 700 nm to 1100 nm in wavelength, and a plurality of light projecting devices—including at least one vertical cavity surface emitting laser (VCSEL)—to determine whether a face presented to the device is human skin with three-dimensional geometry. The system projects a light pattern onto the scene, captures images of the reflected light, and performs independent checks, including detecting a face in the captured image, determining a material classification by analyzing the distribution of light intensity reflected from the face region and comparing it against one or more reference images, generating a depth map based on depth differences in the reflected light pattern, and authenticating the face based on the detection of the face, the classification of the face, and the depth map.

D. The '760 Patent

1. Identification and Ownership of the '760 Patent

44. trinamiX GmbH owns by assignment the entire right, title, and interest in and to the '760 patent, titled "Optical skin detection for face unlock," which was duly issued by the United States Patent and Trademark Office on July 15, 2025, and names Peter Schillen, Benjamin Guthier, Friedrich Schick, Manuel Guenther, Lars Diesselberg, and Christian Lennartz as co-inventors. The

'760 patent issued from U.S. Patent Application Serial No. 18/264,326, filed on August 4, 2023. The application is a U.S. National Phase Application of International Patent Application No. PCT/EP22/053941, filed Feb. 17, 2022, which claims priority to EP patent application Ser. No. 21/157,800.0, filed Feb. 18, 2021, the entire contents of which are hereby incorporated by reference herein. The expiration date of the '760 patent is February 17, 2042.

45. A copy of the '760 patent is attached as Exhibit 7. A copy of the assignment from the named inventors to trinamiX GmbH is attached as Exhibit 8. A copy of the prosecution history of the '760 patent is included as Appendix D.

2. Foreign Counterparts to the '760 Patent

46. Exhibit 16 lists each foreign patent and each pending foreign patent application (not already issued as a patent), and each foreign patent application that has been denied, abandoned, or withdrawn, corresponding to the '760 patent, with an indication of the prosecution status of each such patent application. No other foreign patents or patent applications corresponding to the '760 patent have been filed, abandoned, withdrawn, or rejected.

3. Non-Technical Description of the '760 Patent

47. Mobile and other electronic devices may incorporate biometric authentication techniques, such as face authentication. Such face authentication must be secure and be able to protect against attempts to unlock the device by unauthorized persons. Such persons may employ various spoofing techniques that attempt to mimic an authorized user's face, including photographs, masks, or other spoofs that resemble a user's face. It is thus desirable to provide a face authentication technique that can provide high levels of security and protection against a wide variety of spoofs.

48. The '760 patent is directed to a method for face authentication that combines face detection, skin detection, and three-dimensional detection to verify that a face presented to a device

is human skin rather than a spoof. The method performs at least one face detection step that detects a face in a two-dimensional image by identifying predefined geometrical features characteristic of faces. It performs at least one skin detection step that comprises projecting an illumination pattern onto the scene and capturing an image of the reflected light. The system analyzes the beam profiles of the reflection features within the detected face region to determine at least one material property and characterizes the face as skin if that material property corresponds to a property characteristic of skin. At least one 3D detection step is performed, in which the system analyzes beam profiles of reflection features within the detected face region to determine depth information, and characterizes the face as a three-dimensional object if the depth level deviates from the expected depth level of a pre-determined or pre-defined depth level of plane objects. Authentication is granted if the detected face is characterized as skin and the detected face is characterized as a 3D object.

E. The '394 Patent

1. Identification and Ownership of the '394 Patent

49. trinamiX GmbH owns by assignment the entire right, title, and interest in and to the '394 patent, titled "Detector for identifying at least one material property," which was duly issued by the United States Patent and Trademark Office on May 13, 2025, and names Friedrich Schick, Peter Schillen, Patrick Schindler, Andre Schmidt, Michael Eberspach, Christian Lennartz, Robert Send, Lars Diesselberg, Heiko Hengen, Ingmar Bruder, Jakob Unger, and Christian Bonsignore as co-inventors. The '394 patent issued from U.S. Patent Application Serial No. 18/425,144, filed on January 29, 2024. This application is a continuation application of U.S. Patent Application Serial No. 17/439,492, filed September 15, 2021, which issued as U.S. Patent No. 11,947,013, and which is a U.S. National Phase Application of International Patent Application No. PCT/EP2020/056759, filed March 13, 2020, which claims priority to European Patent Application No. 19163250.4, filed

March 15, 2019, the entire contents of which are hereby incorporated by reference herein. The expiration date of the '394 patent is March 13, 2040.

50. A copy of the '394 patent is attached as Exhibit 9. A certified copy of the assignment from the named inventors to trinamiX GmbH is attached as Exhibit 10. A certified copy of the prosecution history of the '394 patent is included as Appendix E.

2. Foreign Counterparts to the '394 Patent

51. Exhibit 16 lists each foreign patent and each pending foreign patent application (not already issued as a patent), and each foreign patent application that has been denied, abandoned, or withdrawn, corresponding to the '394 patent, with an indication of the prosecution status of each such patent application. No other foreign patents or patent applications corresponding to the '394 patent have been filed, abandoned, withdrawn, or rejected.

3. Non-Technical Description of the '394 Patent

52. Various recognition systems increasingly need to identify what a surface is made of (e.g., to distinguish human skin from a photograph, mask, or other inanimate surface, or to distinguish tissue, fabric, and other materials from one another). Material classification was conventionally performed by shining light through a sample and analyzing extinction, which requires access to both sides of the sample. It was thus desirable to provide a material identification technique that could reliably identify a material from a beam profile of light reflected off an object.

53. The '394 patent is directed to a system and instructions for classifying an object from the intensity distribution of structured light reflected from it. A structured light projector emits a plurality of light beams toward an object. A semiconductor detector sensitive to light receives light reflected from the object and outputs an image characterizing the intensity of that reflected light. A processor then selects at least one region of the image characterizing the reflected light and determines the distribution of light intensity of a first subregion inclusive of a geometric

center of the region and the distribution of light intensity of a second subregion extending from the first subregion. The processor compares the distributions of light intensity of the first and second subregions to a distribution of light intensity of a corresponding region in reference image data derived from a plurality of reference images, determines a classification of the object based on that comparison, and outputs a signal comprising information related to the classification of the object.

F. The '352 Patent

1. Identification and Ownership of the '352 Patent

54. trinamiX GmbH owns by assignment the entire right, title, and interest in and to the '352 patent, titled "Detector for identifying at least one material property," which was duly issued by the United States Patent and Trademark Office on June 17, 2025, and names Friedrich Schick, Peter Schillen, Patrick Schindler, Andre Schmidt, Michael Eberspach, Christian Lennartz, Robert Send, Lars Diesselberg, Heiko Hengen, Ingmar Bruder, Jakob Unger, and Christian Bonsignore as co-inventors. The '352 patent issued from U.S. Patent Application Serial No. 18/743,352, filed on June 14, 2024. This application is a continuation of U.S. Patent Application Serial No. 18/425,144, filed on January 29, 2024, which is a continuation application of U.S. Patent Application Serial No. 17/439,492, filed September 15, 2021, which issued as U.S. Patent No. 11,947,013, and which is a U.S. National Phase Application of International Patent Application No. PCT/EP2020/056759, filed March 13, 2020, which claims priority to European Patent Application No. 19163250.4, filed March 15, 2019, the entire contents of which are hereby incorporated by reference herein. The expiration date of the '352 patent is March 13, 2040.

55. A copy of the '352 patent is attached as Exhibit 11. A copy of the assignment from the named inventors to trinamiX GmbH is attached as Exhibit 12. A copy of the prosecution history of the '352 patent is included as Appendix F.

2. Foreign Counterparts to the '352 Patent

56. Exhibit 16 lists each foreign patent and each pending foreign patent application (not already issued as a patent), and each foreign patent application that has been denied, abandoned, or withdrawn, corresponding to the '352 patent, with an indication of the prosecution status of each such patent application. No other foreign patents or patent applications corresponding to the '394 patent have been filed, abandoned, withdrawn, or rejected.

3. Non-Technical Description of the '352 Patent

57. Various recognition systems increasingly need to identify what a surface is made of (*e.g.*, to distinguish human skin from a photograph, mask, or other inanimate surface, or to distinguish tissue, fabric, and other materials from one another). Material classification was conventionally performed by shining light through a sample and analyzing extinction, which requires access to both sides of the sample. It was thus desirable to provide a material identification technique that could reliably identify a material from a beam profile of light reflected off an object.

58. The '352 patent is directed to a system and instructions for classifying an object by evaluating a material feature. A light projector emits an illumination pattern for illumination of the object. At least one camera comprising a matrix of optical sensors records at least one reflection image. A processor then determines the material property by evaluation of at least one beam profile of the reflection image, applying at least one material dependent image filter.

VI. LICENSEES TO THE ASSERTED PATENTS

59. trinamiX has not licensed the Asserted Patents to Apple or any other third parties.

VII. RESPONDENT'S INFRINGEMENT OF THE ASSERTED PATENTS

60. As discussed above, the Accused Products are iPhones and iPads incorporating face authentication, 3D-sensing, and/or material detection capabilities that infringe the Asserted Patents and are manufactured abroad by or for Respondent, offered for sale and/or sold for importation

into the United States, imported into the United States by or for Respondent, and/or offered for sale or sold within the United States after importation by or for Respondent.

61. Samples of the Accused Products are available on request.

A. Direct Infringement

1. '925 Patent

62. On information and belief, Apple imports, sells for importation, and/or sells after importation into the United States, Accused Products, such as iPhones and iPads incorporating face authentication and components thereof, that directly infringe the '925 patent, literally or under the doctrine of equivalents.

63. Pursuant to Commission Rule 210.12(a)(9)(viii), an exemplary claim chart comparing independent claims 1 and 17 of the '925 patent to the representative articles is attached as Exhibit 17.

2. '421 Patent

64. On information and belief, Apple imports, sells for importation, and/or sells after importation into the United States, Accused Products, such as iPhones and iPads incorporating face authentication and components thereof, that directly infringe the '421 patent, literally or under the doctrine of equivalents.

65. Pursuant to Commission Rule 210.12(a)(9)(viii), an exemplary claim chart comparing independent claim 1 of the '421 patent to the representative articles is attached as Exhibit 18.

3. '334 Patent

66. On information and belief, Apple imports, sells for importation, and/or sells after importation into the United States, Accused Products, such as iPhones and iPads incorporating

face authentication and components thereof, that directly infringe the '334 patent, literally or under the doctrine of equivalents.

67. Pursuant to Commission Rule 210.12(a)(9)(viii), an exemplary claim chart comparing independent claim 1 of the '334 patent to the representative articles is attached as Exhibit 19.

4. '760 Patent

68. On information and belief, Apple imports, sells for importation, and/or sells after importation into the United States, Accused Products, such as iPhones and iPads incorporating face authentication and components thereof, that directly infringe the '760 patent, literally or under the doctrine of equivalents.

69. Pursuant to Commission Rule 210.12(a)(9)(viii), an exemplary claim chart comparing independent claim 1 of the '760 patent to the representative articles is attached as Exhibit 20.

5. '394 Patent

70. On information and belief, Apple imports, sells for importation, and/or sells after importation into the United States, Accused Products, such as iPhones and iPads incorporating face authentication and components thereof, that directly infringe the '394 patent, literally or under the doctrine of equivalents.

71. Pursuant to Commission Rule 210.12(a)(9)(viii), an exemplary claim chart comparing independent claims 1 and 15 of the '394 patent to the representative articles is attached as Exhibit 21.

6. '352 Patent

72. On information and belief, Apple imports, sells for importation, and/or sells after importation into the United States, Accused Products, such as iPhones and iPads incorporating

face authentication and components thereof, that directly infringe the '352 patent, literally or under the doctrine of equivalents.

73. Pursuant to Commission Rule 210.12(a)(9)(viii), an exemplary claim chart comparing independent claim 1 of the '352 patent to the representative articles is attached as Exhibit 22.

B. Indirect Infringement

74. On information and belief, Apple both induces infringement of the Asserted Patents under 35 U.S.C. § 271(b) and contributes to the infringement of the Asserted Patents under 35 U.S.C. § 271(c).

75. Apple has had knowledge of the Asserted Patents, and its infringement of the Asserted Patents, since at least the filing of this Complaint and of the complaint filed in the U.S. District Court for the Western District of Texas alleging infringement of the Asserted Patents. Apple also knows that the acts it induces constitute infringement of the Asserted Patents. Despite this knowledge of the Asserted Patents, Apple continues to actively encourage and instruct its distributors, customers, and end users to use the Accused Products in ways that directly infringe the Asserted Patents.

76. On information and belief, Apple's distributors, customers, and end users directly infringe Asserted Patents by, for example, using, offering to sell, and selling within the United States, and importing into the United States, without authority or license, the Accused Products. On information and belief, Apple actively, knowingly, and intentionally has induced infringement of the Asserted Patents by its distributors, customers, and end users by, for example, controlling the design and manufacture of, offering for sale, selling, supplying, and otherwise providing instruction and guidance regarding the above-described products with the knowledge and specific intent to encourage and facilitate infringing uses of such products by its customers both inside and

outside the United States. For example, Apple publicly provides documentation, including datasheets and manuals instructing customers on uses of Apple's products that infringe the Asserted Patents.³

77. Apple induces infringement of the Asserted Patents. Apple knows of the Asserted Patents, knows that the acts it encourages constitute infringement of the Asserted Patents, and specifically intends that its customers and end users perform those acts. Apple designs the Accused Products so that face authentication operates in the infringing manner by default, instructs its customers and end users to enroll a face and to use Face ID, and provides user guides, support pages, and technical documentation directing them to do so. Apple's customers and end users directly infringe the Asserted Patents when they use the Accused Products as Apple instructs. *See* Exhibits 17-23.

78. Apple contributes to the infringement of the Asserted Patents. Apple imports, sells for importation, and sells after importation the Accused Products, including the face authentication hardware and software components thereof. Those components are a material part of the claimed inventions, are especially made or especially adapted for use in an infringement of the Asserted Patents, and are not staple articles or commodities of commerce suitable for substantial non-infringing use. The face authentication subsystem of the Accused Products, including its illumination, near-infrared imaging, material classification, and depth determination functions, has no substantial use other than to practice the Asserted Patents. Apple makes and supplies those components with knowledge of the Asserted Patents and of that infringing use.

³ *See, e.g.,* Exhibit 23.

VIII. SPECIFIC INSTANCES OF UNFAIR IMPORTATION AND SALE

79. On information and belief, the Accused Products are manufactured abroad and then sold for importation into the United States by Respondent or on its behalf, imported into the United States by Respondent or on its behalf, and/or sold after importation by Respondent or on its behalf.

80. For example, Apple imports or sells for importation the iPhone 17 that infringes each of the Asserted Patents. Complainants purchased an iPhone 17 from the Apple store at 189 The Grove Drive Los Angeles, CA 90036 on August 23, 2026. The marking on the product indicates that it was assembled in India. *See* Exhibit 24.

81. As another example, Apple imports or sells for importation the iPad Pro 11-inch (M5), which infringes each of the Asserted Patents. Complainants ordered an iPad Pro 11-inch (M5) from the Apple store at 189 The Grove Drive Los Angeles, CA 90036 on August 23, 2026. The marking on the product indicates that it was assembled in Vietnam. *See* Exhibit 24.

IX. HARMONIZED TARIFF SCHEDULE NUMBERS

82. The Accused Products are classified under at least the following subheadings of the Harmonized Tariff Schedule of the United States: 8517.13.0000 and 8471.30.0100. These classifications are exemplary in nature and not intended to restrict the scope of any exclusion order or other remedy ordered by the Commission.

X. THE DOMESTIC INDUSTRY RELATING TO THE ASSERTED PATENTS

83. A domestic industry, as set forth in 19 U.S.C. § 1337(a)(2) and defined by 19 U.S.C. § 1337(a)(3), exists, or alternatively is in the process of being established, in the United States in relation to trinamiX products that are protected by the Asserted Patents. To the extent it is determined that a domestic industry does not currently exist with respect to any Asserted Patent, trinamiX is in the process of establishing a domestic industry with respect to that patent, and there

is a significant likelihood that trinamiX will satisfy the economic domestic industry requirement in the future, as set forth below.

A. Technical Prong

84. The Domestic Industry Products are trinamiX's face authentication hardware and software, which includes trinamiX's face authentication software libraries and software development kit (SDK) and related hardware.

85. Representative claim charts, attached as Exhibits 25C-30C, show that the Domestic Industry Products are protected by at least one claim of each of the Asserted Patents.⁴

86. Samples of the Domestic Industry Products are available on request.

B. Economic Prong

87. For all Asserted Patents, there is a domestic industry pursuant to 19 U.S.C. § 1337(a)(3)(A), (B), and (C) based on trinamiX's continuing significant U.S. investment in plant and equipment, and labor or capital, as well as trinamiX's continuing substantial U.S. investment in research and development. Details regarding these investments are set forth in the Confidential Declaration of Uwe Schneider (Confidential Exhibit 31C).

88. The domestic industry is based on significant investments trinamiX has made, and/or plans to make, and activities trinamiX has undertaken, and/or plans to undertake, in the United States relating to products protected by one or more claims of the Asserted Patents. Specific, non-limiting examples of these investments and activities are set forth in the Confidential Declaration of Uwe Schneider (Confidential Exhibit 31C).

⁴ The Domestic Industry Products are protected by additional claims of the Asserted Patents, and Complainants may establish the technical prong of the domestic industry requirement through claims other than those explicitly charted.

89. trinamiX Sensing LLC was established in 2025 to build trinamiX's face authentication business in the United States. It leases and operates an office and laboratory space at 7800 Shoal Creek Boulevard, Suite 200N, Austin, Texas, which it has equipped for the development and testing of the Domestic Industry Products. It has hired and employs a dedicated U.S. team of software engineers, hardware engineers, data scientists, and test technicians, who perform engineering, research, and development directed to the face authentication technology claimed in the Asserted Patents. That work includes development and maintenance of the 3D and material sensing software pipeline, material detection improvements, machine learning pipeline development, and hardware setup and testing for the Domestic Industry Products.

90. trinamiX's U.S. investments are significant in the context of trinamiX's business. The Austin facility houses a significant proportion of the personnel involved in trinamiX's biometric imaging business worldwide, and trinamiX Sensing LLC activities are wholly devoted to the Domestic Industry Products. The dollar amounts of trinamiX's investments in plant and equipment, in labor and capital, and in engineering and research and development, together with the bases for their allocation, are set forth in the Confidential Declaration of Uwe Schneider (Confidential Exhibit 31C).

91. trinamiX's domestic investments are significant and substantial under Section 337, both in absolute terms and relative to trinamiX's overall operations, and are vital to trinamiX's domestic business with respect to the Domestic Industry Products and represent significant added value.

92. To the extent it is determined that a domestic industry does not currently exist, trinamiX is in the process of establishing a domestic industry with respect to the Domestic Industry Products because it is actively engaged in tangible and necessary steps leading to the exploitation

of its intellectual property rights and/or the development and commercialization of the Domestic Industry Products, and there is a significant likelihood that an industry will be established in the United States in the future. Specific non-exhaustive examples of the active steps taken by trinamiX to establish a domestic industry are included in the Confidential Declaration of Uwe Schneider (Confidential Exhibit 31C).

93. There is a significant likelihood that trinamiX will satisfy the economic domestic industry requirement in the future. trinamiX has taken tangible steps toward establishing a domestic industry, including forming trinamiX Sensing LLC, registering it to transact business in Texas, leasing and equipping the Austin facility, hiring and onboarding a dedicated U.S. engineering team, and assigning that team to project work directed to the Domestic Industry Products. Those steps are not merely preparatory or speculative. The U.S. team is performing the engineering and research and development work today, the Austin facility is operating today, and trinamiX has committed budgeted funds to that work for the current and following fiscal years. These facts are set forth in further detail in the Confidential Declaration of Uwe Schneider (Confidential Exhibit 31C).

XI. RELATED LITIGATION

94. trinamiX has filed a complaint in the U.S. District Court for the Western District of Texas alleging infringement of the Asserted Patents against Respondent. *trinamiX Sensing LLC and trinamiX GmbH v. Apple Inc.*, Case No. 7:26-cv-00348 (W.D. Tex. Sept. 3, 2006).

95. trinamiX GmbH has also brought an action against Respondent before the Unified Patent Court, Local Division Mannheim, Case No. PR-UPC-CFI-0003171/2026, filed September 3, 2026, asserting European Patent No. 4,295,252, which is a foreign counterpart of the asserted '760, '334, '421, and '925 patents.

96. All conditions precedent to trinamiX's right to the relief requested have been performed, waived, or otherwise satisfied.

XII. RELIEF REQUESTED

WHEREFORE, trinamiX respectfully requests that the Commission:

- (a) Institute an investigation pursuant to Section 337 of the Tariff Act of 1930, as amended, 19 U.S.C. § 1337, with respect to Respondent's violations of that section based on the importation into the United States, sale for importation, and/or sale within the United States after importation of Apple iPhones and iPads that infringe one or more claims of the Asserted Patents;
- (b) Schedule and conduct a hearing pursuant to Section 337(c) for the purpose of receiving evidence and hearing argument concerning whether there has been a violation of Section 337, and following the hearing, determine that there has been such a violation;
- (c) Issue a permanent limited exclusion order pursuant to 19 U.S.C. § 1337(d) forbidding entry into the United States of Apple iPhones and iPads that infringe one or more claims of the Asserted Patents and that are manufactured by, imported by, offered for sale by, sold for importation by, and/or sold after importation by or on behalf of Respondent, its subsidiaries, related companies, and agents;
- (d) Issue permanent cease and desist orders pursuant to 19 U.S.C. § 1337(f) prohibiting Respondent, its domestic subsidiaries, related companies, and agents from importing, selling for importation, marketing, advertising, distributing, offering for sale, selling, using after importation, selling after importation, or otherwise transferring within the United States (except for exportation) Apple iPhones and iPads that infringe one or more claims of the Asserted Patents;

- (e) Impose a bond pursuant to 19 U.S.C. § 1337(j) upon importation of Apple iPhones and iPads that infringe one or more claims of the Asserted Patents during the 60-day Presidential review period; and
- (f) Issue such other and further relief as the Commission deems just and proper under the law, based on the facts determined by the investigation and the authority of the Commission.

Dated: September 3, 2026

Respectfully submitted,

/s/ S. Alex Lasher

S. Alex Lasher

Carl Spilly

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